

Conditions for eigenvalue configurations of two real symmetric matrices: a stochastic definition

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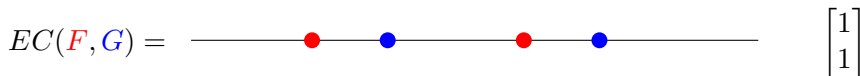
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Out: $\begin{pmatrix} \sigma(D_{00}) \\ \sigma(D_{01}) \\ \sigma(D_{10}) \\ \sigma(D_{11}) \end{pmatrix} \in \left\{ \begin{pmatrix} 1 \\ 2 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 0 \\ 0 \end{pmatrix} \right\}$ where

$$D_{00} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$D_{01} = \begin{bmatrix} -a_{2,2} - a_{1,1} + 2b_{1,1} & 2b_{1,2} \\ 2b_{1,2} & -a_{1,1} - a_{1,1} + 2b_{2,2} \end{bmatrix}$$

$$D_{11} = \begin{bmatrix} a_{1,1}a_{2,2} - a_{1,2}^2 - (a_{1,1} + a_{2,2})b_{1,1} + b_{1,1}^2 + b_{1,2}^2 & -b_{1,2}(a_{1,1} + a_{2,2} - b_{1,1} - b_{2,2}) \\ -b_{1,2}(a_{1,1} + a_{2,2} - b_{1,1} - b_{2,2}) & a_{1,1}a_{2,2} - a_{1,2}^2 - (a_{1,1} + a_{2,2})b_{2,2} + b_{1,2}^2 + b_{2,2}^2 \end{bmatrix}$$

$\vdots$

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- Let F be the adjacency matrix of a graph.
- Some properties are related to the eigenvalues of F .
- How do eigenvalues change when edges are added or removed?
- Changing edges corresponds to adding a matrix U to F
- Hence, one might be interested in the eigenvalue configuration of

$$F \quad \text{and} \quad G = F + U.$$

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In : $\exists_{\alpha_1, \dots, \alpha_m, \beta_1, \dots, \beta_n} \Phi_{\mathbf{c}}(\alpha_1, \dots, \alpha_m, \beta_1, \dots, \beta_n, a_{ij}, b_{ij})$

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Challenge: Develop an algorithm that exploits the problem structure.

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- Let f denote the characteristic polynomial of F .

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Definition (Generic)

Let $F \in \mathbb{R}^{m \times m}$, $G \in \mathbb{R}^{n \times n}$ be symmetric and generic (i.e. having **distinct eigenvalues**).

Then the **generic** eigenvalue configuration $\text{GEC}(F, G)$ is the vector $\mathbf{c} \in \mathbb{N}^{m \times m}$ where

$$\mathbf{c}_t = \#\{j : \beta_j \in (\alpha_t, \alpha_{t+1})\}.$$

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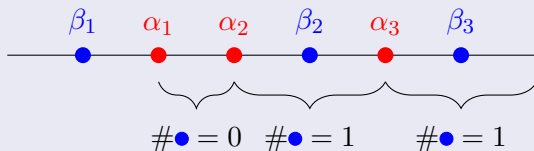
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Example



$$\text{GEC}(F, G) = (0, 1, 1).$$

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① $C_{\text{sig}} = V H^{-1} \in \mathbb{N}^{m \times 2^m}$, where

$$V_{t,s} = \delta_{\{v(s,+), m-t\}}$$

$$H_{e,s} = s_1^{e_1} \cdots s_m^{e_m}$$

$$t \in \{1, \dots, m\}, \quad s \in \{-, +\}^m$$

$$e \in \{0, 1\}^m, \quad s \in \{-, +\}^m$$

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② $A_{\text{sig}}(\mathbf{F}, \mathbf{G}) \in \mathbb{Z}^{2^m}$, where

$$\begin{aligned} (A_{\text{sig}})_e &= \sigma(f_e(\mathbf{G})) & e &\in \{0, 1\}^m \\ f_e &= f^{(0)e_0} \cdots f^{(m-1)e_{m-1}} & f^{(k)} &= k\text{-th derivative} \\ f &= \det(xI_m - \mathbf{F}). \end{aligned}$$

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Remark: Since all the eigenvalues are real, one can easily express the signature in terms of sign variation count and the entries of $\mathbf{F}$ and $\mathbf{G}$.

Generic Case: Proof Outline

Sketch of Proof: (with full details: ≈ 24 pages in [22])

$$\begin{aligned} \mathbf{c} &= \text{GEC}(F, G) \\ &\Updownarrow \\ \mathbf{c}_t &= \#\{j : \alpha_t < \beta_j < \alpha_{t+1}\} \\ &\Updownarrow \\ \mathbf{c}_t &= \#\{j : \overline{\#\{x : f(x) = 0 \wedge x > \beta_j\}} = m - t\} \\ &\Updownarrow \\ \mathbf{c} &= V \begin{bmatrix} \#\{j : \text{sign}(f^{(0)}(\beta_j), \dots, f^{(m-1)}(\beta_j)) = - \dots -\} \\ \vdots \end{bmatrix} \\ &\Updownarrow \\ \mathbf{c} &= V H^{-1} \begin{bmatrix} \sigma \left((f^{(0)})^0 \dots (f^{(m-1)})^0 (G) \right) \\ \vdots \end{bmatrix} \\ &\Updownarrow \\ \mathbf{c} &= C_{\text{sig}} A_{\text{sig}}(F, G). \end{aligned}$$

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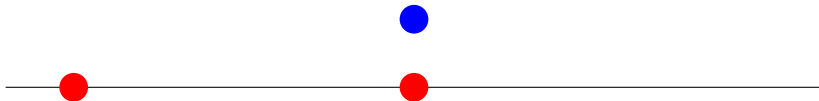
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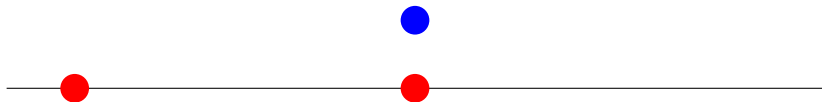
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Shared Eigenvalues: Idea



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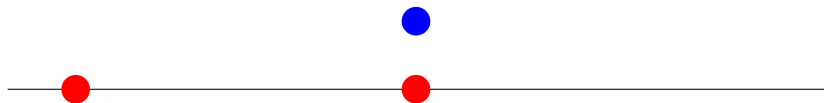


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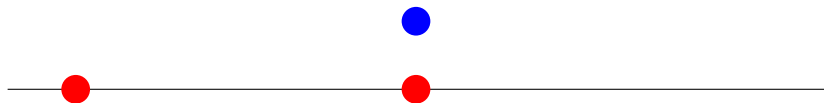


Q: What do we get when we run the algorithm?

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$$\text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}) = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$

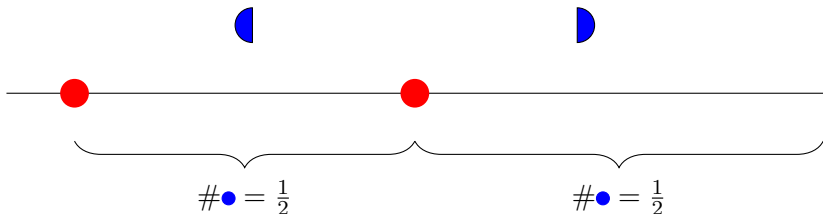
Shared Eigenvalues: Idea



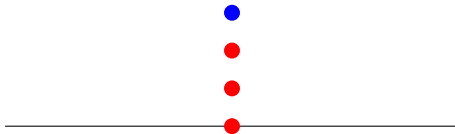
Q: What do we get when we run the algorithm?

A:

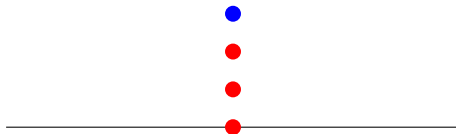
$$\text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}) = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$



Shared Eigenvalues: Idea



Shared Eigenvalues: Idea



$$\text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}) = \begin{bmatrix} 3/8 \\ 3/8 \\ 1/8 \end{bmatrix}$$

Shared Eigenvalues: New Definition

Definition (Generalized Eigenvalue Configuration)

Let F and G be arbitrary real symmetric matrices. Then

$$\begin{aligned}\text{EC}(F, G) &= \mathbb{E} \text{GEC}(F_d, G) \\ &= \frac{1}{2^m} \sum_{d \in \{-\varepsilon, \varepsilon\}^m} \text{GEC}(F_d, G)\end{aligned}$$

where

$d \in \{-\varepsilon, \varepsilon\}^m$ i.i.d. random variables

F_d = matrix with eigenvalues $\alpha_i + d_i$

ε = "small enough" positive number.

Shared Eigenvalues: New Definition

Definition

$$\begin{aligned}\text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}) &= \mathbb{E} \text{ GEC}(\textcolor{red}{F}_d, \textcolor{blue}{G}) \\ &= \frac{1}{2^{\textcolor{red}{m}}} \sum_{d \in \{-\varepsilon, \varepsilon\}^{\textcolor{red}{m}}} \text{GEC}(\textcolor{red}{F}_d, \textcolor{blue}{G})\end{aligned}$$

Some remarks:

Shared Eigenvalues: New Definition

Definition

$$\begin{aligned}\text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}) &= \mathbb{E} \text{ GEC}(\textcolor{red}{F}_d, \textcolor{blue}{G}) \\ &= \frac{1}{2^{\textcolor{red}{m}}} \sum_{d \in \{-\varepsilon, \varepsilon\}^{\textcolor{red}{m}}} \text{GEC}(\textcolor{red}{F}_d, \textcolor{blue}{G})\end{aligned}$$

Some remarks:

- ε chosen so that $\textcolor{red}{F}_d$ and $\textcolor{blue}{G}$ do not share eigenvalues.

Shared Eigenvalues: New Definition

Definition

$$\begin{aligned}\text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}) &= \mathbb{E} \text{GEC}(\textcolor{red}{F}_d, \textcolor{blue}{G}) \\ &= \frac{1}{2^{\textcolor{red}{m}}} \sum_{d \in \{-\varepsilon, \varepsilon\}^{\textcolor{red}{m}}} \text{GEC}(\textcolor{red}{F}_d, \textcolor{blue}{G})\end{aligned}$$

Some remarks:

- ε chosen so that $\textcolor{red}{F}_d$ and $\textcolor{blue}{G}$ do not share eigenvalues.
- ε chosen so that perturbed red points do not move too far.

Shared Eigenvalues: New Definition

Definition

$$\begin{aligned}\text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}) &= \mathbb{E} \text{GEC}(\textcolor{red}{F}_d, \textcolor{blue}{G}) \\ &= \frac{1}{2^{\textcolor{red}{m}}} \sum_{d \in \{-\varepsilon, \varepsilon\}^{\textcolor{red}{m}}} \text{GEC}(\textcolor{red}{F}_d, \textcolor{blue}{G})\end{aligned}$$

Some remarks:

- ε chosen so that $\textcolor{red}{F}_d$ and $\textcolor{blue}{G}$ do not share eigenvalues.
- ε chosen so that perturbed red points do not move too far.
- New definition still works when $\textcolor{red}{F}$ and $\textcolor{blue}{G}$ do not share eigenvalues.

Shared Eigenvalues: Example

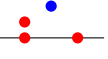
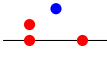
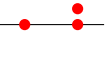
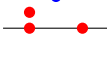
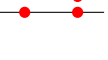


Shared Eigenvalues: Example

d	c	EC	d	c	EC
— — —		$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	+ — —		$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$
— — +		$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$	+ — +		$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
— + —		$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$	+ + —		$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
— + +		$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$	+ + +		$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Shared Eigenvalues: Example



d	c	EC	d	c	EC
— — —		$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	+ — —		$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$
— — +		$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$	+ — +		$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
— + —		$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$	+ + —		$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
— + +		$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$	+ + +		$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}) = \frac{1}{8} \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 3/8 \\ 3/8 \\ 1/8 \end{bmatrix}$$

Shared Eigenvalues: Proof Outline

Sketch of Proof: (with full details: ≈ 11 pages in [22])

$$\begin{array}{ccc} \text{EC}(\textcolor{red}{F}, \textcolor{blue}{G}) & \stackrel{?}{=} & C_{\text{sig}} A_{\text{sig}}(\textcolor{red}{F}, \textcolor{blue}{G}) \\ \text{decompose} \quad \parallel & & \parallel \quad \text{combine} \\ \sum_{j=1}^n \text{EC}(\textcolor{red}{F}, [\textcolor{blue}{\beta}_j]) & = & \sum_{j=1}^n C_{\text{sig}} A_{\text{sig}}(\textcolor{red}{F}, [\textcolor{blue}{\beta}_j]) \end{array}$$

Contents

1 Problem

2 Generic Case

3 Shared Eigenvalues Case

4 Future Work

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Short-term:

- Find applications

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- Generalize to tensors

Future Work

Short-term:

- Find applications
- Prune redundancies from output

Long-term:

- Generalize to > 2 matrices
- Generalize to tensors

Other solutions:

- We have another algorithm for GEC which uses the theory of symmetric polynomials [23].
- Can we find more?

References I



R. Descartes.

Géométrie.

1636.

In *A source book in Mathematics*, pages 90-131. Harvard University Press, 1969.



M. L. Telek.

Geometry of the signed support of a multivariate polynomial and descartes' rule of signs.

SIAM Journal on Applied Algebra and Geometry, 8(4):968–1000, 2024.



Bernhard Beckermann, Daniel Kressner, and Marcel Schweitzer.

Low-rank updates of matrix functions.

SIAM Journal on Matrix Analysis and Applications, 39(1):539–565, 2018.



A. Tarski.

The completeness of elementary algebra and geometry.

1930.

Reprinted in 1967.

References II



G. E. Collins.

Quantifier elimination for the elementary theory of real closed fields by cylindrical algebraic decomposition.

In *Lecture Notes In Computer Science*, pages 134–183. Springer-Verlag, Berlin, 1975.

Vol. 33.



M. Ben-Or, D. Kozen, and J. H. Reif.

The complexity of elementary algebra and geometry.

J. Comput. System Sci., 32(2):251–264, 1986.



D. Yu. Grigor'ev.

The complexity of deciding Tarski algebra.

Journal of Symbolic Computation, 5(1,2):65–108, 1988.



L. Gonzalez-Vega, H. Lombardi, T. Recio, and M.-F. Roy.

Sturm-Habicht sequences.

In *Proceedings of the ACM-SIGSAM 1989 International Symposium on Symbolic and Algebraic Computation*, pages 136–146, July 1989.

References III



H. Hong.

An improvement of the projection operator in cylindrical algebraic decomposition.
In International Symposium of Symbolic and Algebraic Computation (ISSAC-90),
pages 261–264. ACM, 1990.



H. Hong.

Improvements in CAD-based Quantifier Elimination.
PhD thesis, The Ohio State University, 1990.



G. E. Collins and H. Hong.

Partial cylindrical algebraic decomposition for quantifier elimination.
Journal of Symbolic Computation, 12(3):299–328, sep 1991.



H. Hong.

Simple solution formula construction in cylindrical algebraic decomposition based
quantifier elimination.
In International Conference on Symbolic and Algebraic Computation ISSAC-92,
pages 177–188, 1992.

References IV



J. Renegar.

On the computational complexity and geometry of the first-order theory of the reals.

Journal of Symbolic Computation, 13(3):255–352, 1992.



J. F. Canny.

Improved algorithms for sign and existential quantifier elimination.

Computer Journal, 36:409–418, 1993.

In a special issue on *computational quantifier elimination*, edited by H. Hong.



V. Weispfenning.

Quantifier elimination for real algebra – the cubic case.

In *International Symposium on Symbolic and Algebraic Computation 94*, pages 258–263, 1994.

References V



L. Gonzalez-Vega.

A combinatorial algorithm solving some quantifier elimination problems.

In B. Caviness and J. Johnson, editors, *Quantifier Elimination and Cylindrical Algebraic Decomposition*. Springer Verlag, 1996.

Texts and Monographs in Symbolic Computation.



S. McCallum.

Factors of iterated resultants and discriminants.

J. Symb. Comput., 27(4):367–385, 1999.



C. W. Brown.

Improved projection for cylindrical algebraic decomposition.

J. Symb. Comput., 32(5):447–465, 2001.



A. Strzebonski.

Cylindrical algebraic decomposition using validated numerics.

J. Symb. Comput., 41(9):1021–1038, 2006.

References VI



H. Hong and M. S. El Din.

Variant quantifier elimination.

J. Symb. Comput., 47(7):883–901, 2012.



Rizeng Chen.

The geometry of cylindrical algebraic decomposition.

ACM Commun. Comput. Algebra, 58(3):67–71, February 2025.



Hoon Hong, Daniel Profili, and J Rafael Sendra.

Conditions for eigenvalue configurations of two real symmetric matrices: (signature approach).

arXiv preprint arXiv:2401.00866, 2025.



Hoon Hong, Daniel Profili, and J Rafael Sendra.

Conditions for eigenvalue configurations of two real symmetric matrices (symmetric polynomial approach).

arXiv preprint arXiv:2401.00869, 2025.

Thank you!